

Criterion B: Developing ideas

ii. Develop a range of feasible design ideas

Before drawing anything, I wrote down the one question every idea had to answer: how does a student who has just walked out of a hard test get from 'I feel bad' to one thing that might actually help, without ever losing sight of the counsellor? I came up with four different answers. They are not the same app in four colour schemes. Each idea moves the student through the check-in in a different way, because my research in Criterion A showed that speed and being able to find help were what mattered most to students.

To keep the comparison fair, all four ideas use the same five moods, the same ten tags and the same library of twelve coping cards, and all four could be built for free in Google AI Studio in about 20 hours. I drew every screen at real phone size (360 px wide). The fully annotated boards are in my Canva design; under each board here I evaluate the idea against my B.i specifications. A specification only counts as met if the board shows it, and the score for each category is the number of its specifications that are met.

What the moods and tags mean

Every idea starts from the same check-in, so I defined each mood and tag before designing any screens. 'Bad' means different things to different people, which would make the matching unreliable, so each mood has one plain sentence that appears when it is picked. Each tag has a meaning and a school example, which I used when testing the ideas.

Mood and tag guide

What each mood and tag means, with a real school example, and what the app does with it.

Awful

value 1

"Too much right now. Hard to think about anything else."

EXAMPLE You've felt panicky since lunch and can't take in anything the teacher says.

WHAT THE APP DOES
The counsellor card is shown first, then the matched cards.

Bad

value 2

"Things feel heavy, but you can still get through the lesson."

EXAMPLE You just walked out of a maths test you think went badly and your chest feels tight.

WHAT THE APP DOES
Normal matching. With no feeling tag, it uses the 'low' row.

Okay

value 3

"Not great, not terrible. A bit on edge."

EXAMPLE Your IA draft is due Friday and it keeps popping into your head in class.

WHAT THE APP DOES
Normal matching.

Good

value 4

"Mostly fine. Just checking in."

EXAMPLE A normal day, but you have a presentation next period.

WHAT THE APP DOES
Normal matching. With no feeling tag, it uses the 'no feeling tag' row.

Great

value 5

"Feeling steady and want to keep it that way."

EXAMPLE You finished a big deadline and want a quick reset before the next lesson.

WHAT THE APP DOES
Normal matching, usually a short break or breathing card.

"How I feel" tags

pick any; the first one picks the table row

anxious	Worried about what might happen. Racing heart, butterflies, tight chest. E.G. Waiting outside the exam room.
overwhelmed	Too many things at once. You don't know where to start. E.G. Three deadlines in one week, plus an IA draft.
frustrated	Annoyed or stuck because something isn't working. E.G. Your code won't run and the lesson ends in 10 minutes.
low	Flat, sad or unmotivated. E.G. You've felt down all week and don't want to talk in class.
tired	Drained, sleepy or foggy. E.G. You stayed up late revising and nothing is going in.

"What's happening" tags

pick any; the first one picks the table column

test or exam	Before, during a break in, or just after an assessment. E.G. Mocks, unit tests, quizzes, orals. column: Test
IA or deadline	Coursework or anything with a due date. E.G. IA, EE, TOK essay, MYP Personal Project. column: IA
in class	You're in a lesson and can't leave or talk. Only silent cards are shown. E.G. Sitting in chemistry and feeling it build. column: Class
friends or people	Something with friends, classmates or family. E.G. A falling-out with a friend at break. column: Other
home	Something at home or outside school. E.G. An argument at home before school. column: Other

Moods one required, shown with its sentence on the check-in screen

Tags optional, so a student who can't name the feeling still gets a card

All words stored in content.json

Figure B1: Mood and tag guide (Canva).

Design 1: All-in-One Dashboard

Design 1 came from Daylio, one of the apps I analysed in Criterion A. The best thing about Daylio is that logging a mood takes seconds, because everything is on one screen. I wanted to see what would happen if Pause worked the same way, with no submit button at all: the suggestion card halfway down the page changes as soon as a tag is tapped.

I tested it with a scenario. A DP1 student is waiting outside a chemistry test. They drag the slider towards 'Bad', tap 'anxious' and 'test or exam', and 'Nerves can help' appears straight away. Two taps makes it the fastest of

the four. The problems only showed up once I drew it at real size: fitting everything on one page pushed the counsellor link about 1180 px down, past the student's own history.

Design 1: All-in-One Dashboard

One long scrolling page: mood slider, tags, note, a live-updating suggestion, history, and the counsellor link at the very bottom. Nothing to submit; the card changes as you tap.

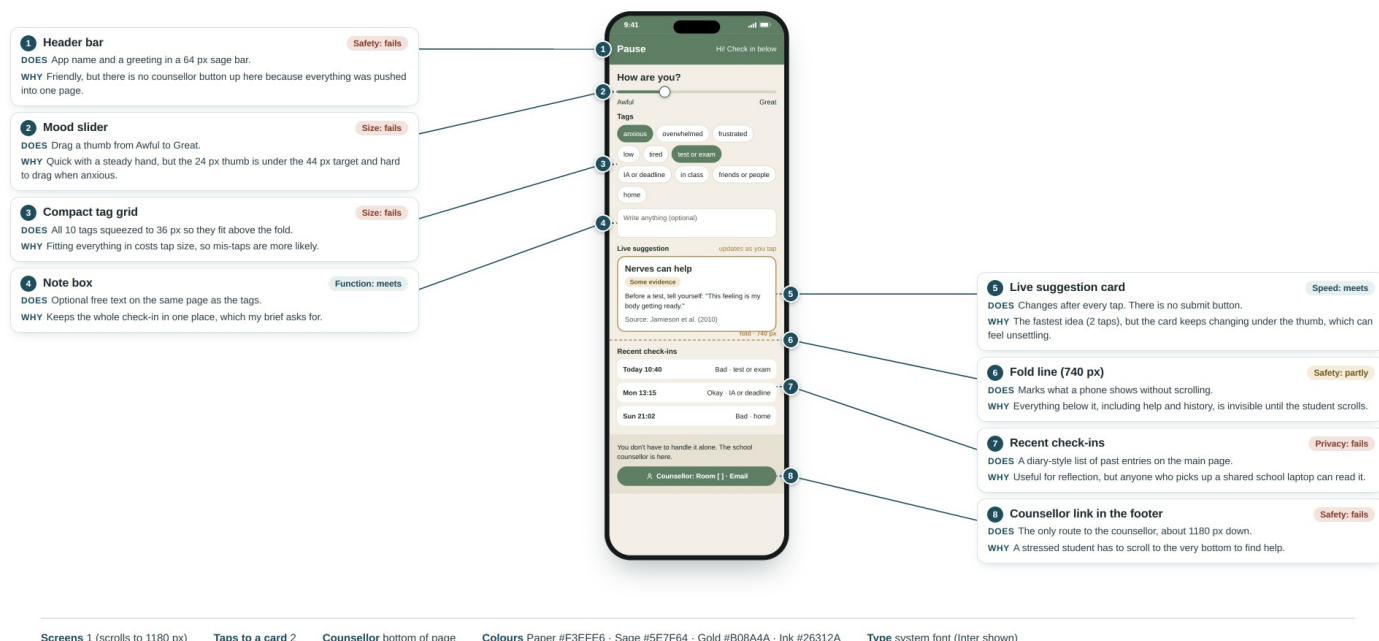


Figure B2: Design 1: All-in-One Dashboard (Canva).

ACCESSFMM	How this design meets my B.i specifications	Score
Aesthetics	Muted sage and paper colours used the same way throughout, so it looks calm.	1/1
Cost	Uses only free tools; free for students.	1/1
Customer / Client	All ten tags are there, so it fits MYP and DP. The live card only offers silent techniques when 'in class' is tapped.	2/2
Environment	Text only, so it loads quickly on school Wi-Fi with no downloaded fonts.	2/2
Safety	Fails the counsellor specification because the only link is at the very bottom. The support line is there, but below the fold, so only partly. Fails privacy because past check-ins appear on the main page of a shared laptop.	1/4 (1 partly)
Size	Reflows to 320 px, but the 24 px slider thumb and 36 px chips are under my 44 px tap target.	1/2
Form and Function	Mood, tags and note share one screen. Two taps to a card is the fastest of all four. Each card shows its evidence label.	4/4
Manufacturing	One React page, buildable in 20 h. New cards only need entries in content.json.	2/2
Materials	Free system font and my own SVG icons, all recorded in the asset register.	1/1
Total	15 of 19 specifications met, 1 partly met, 3 not met.	15/19

Design 2: Step-by-Step Guide

Design 2 goes the other way. Instead of showing everything at once, it asks one question per screen, like a short quiz, with a progress bar along the top. I chose this structure because one large question on a calm, nearly empty screen is the easiest thing to read when you are stressed, and its 56 px mood buttons are almost impossible to mis-tap.

The same scenario showed what that calm costs: mood, Next, tag, Next, Skip, Next. That is five or six taps before the student sees anything useful. Stress weakens the working memory needed to follow a sequence of steps (Arnsten, 2009), so the idea that looks calmest is actually the hardest to finish when you are anxious.

Design 2: Step-by-Step Guide

One question per screen, like a short quiz: mood, then tags, then an optional note, then the suggestion. Very clear, but every step costs a tap.

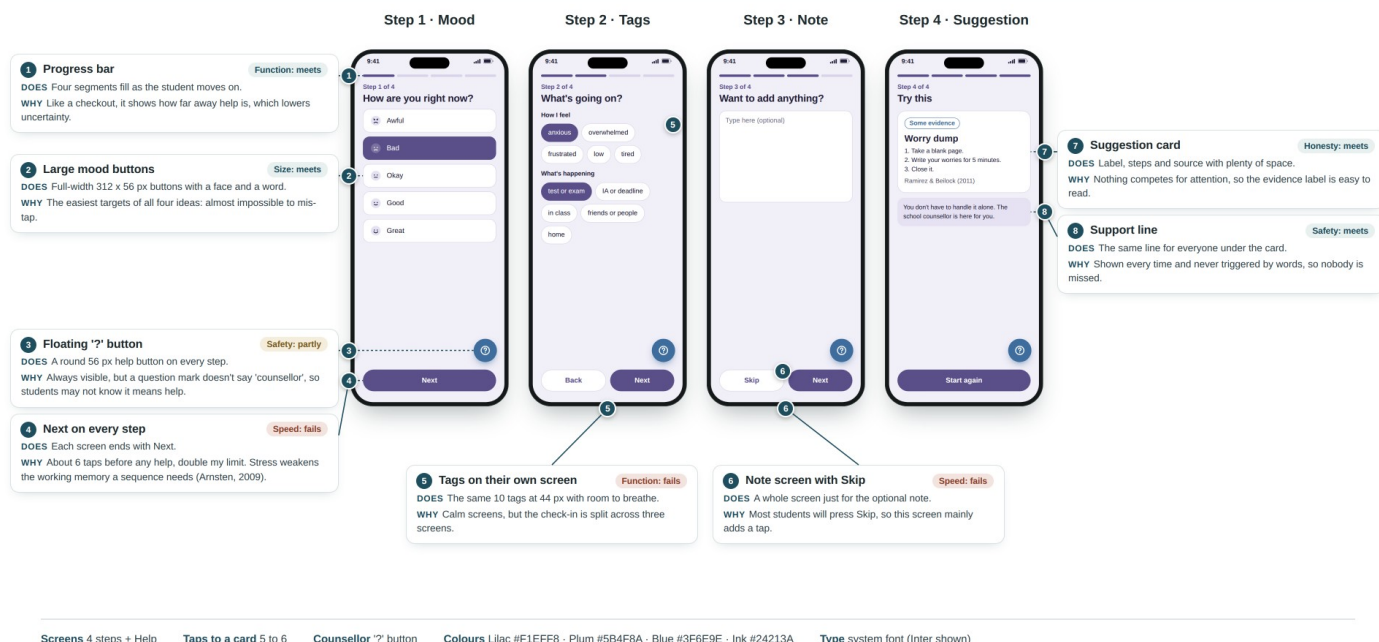


Figure B3: Design 2: Step-by-Step Guide (Canva).

ACCESSFMM	How this design meets my B.i specifications	Score
Aesthetics	Lilac and plum used consistently on every step; the most visually calm of the four.	1/1
Cost	Free to build and use.	1/1
Customer / Client	All ten tags fit on their own screen. Nothing on it would be a problem in class.	2/2
Environment	Text only; each step is a small screen.	2/2
Safety	Help is a '?' button that is always visible but never says who it connects to, so the counsellor specification is only partly met. The support line, privacy and content specifications are all met.	3/4 (1 partly)
Size	Largest tap targets of the four, and nothing under 16 px.	2/2
Form and Function	Fails 'hybrid check-in on one screen' because mood, tags and note are split over three screens. Fails speed with 5 to 6 taps. Matching and evidence labels work as planned.	2/4
Manufacturing	Four simple views, buildable in 20 h. Each new question means a new screen, but cards are still just data.	2/2
Materials	Free and traceable, as for all designs.	1/1
Total	16 of 19 specifications met, 1 partly met, 2 not met.	16/19

Design 3: Scripted Chat (no AI)

Design 3 was inspired by Wysa, whose chat style felt the most friendly of the apps I analysed. I kept the feeling of a conversation but took the AI out completely: every message is written in advance and the student answers by tapping reply chips, so nothing is generated and nothing reads what they type. I expected this to be the most popular idea with younger students, and in the peer vote it was.

Drawing it at real size showed the problems. Only four chips fit across a phone, so 'IA or deadline' and 'home' had to go. The evidence label shrinks to a 12 px grey line. Worst of all, the message box makes it look as if someone is reading, so a student might type something serious and wait for a reply that never comes.

Design 3: Scripted Chat (no AI)

A friendly chat where every question is pre-written and answered by tapping chips. It feels personal, but it looks like someone is listening when nobody is.

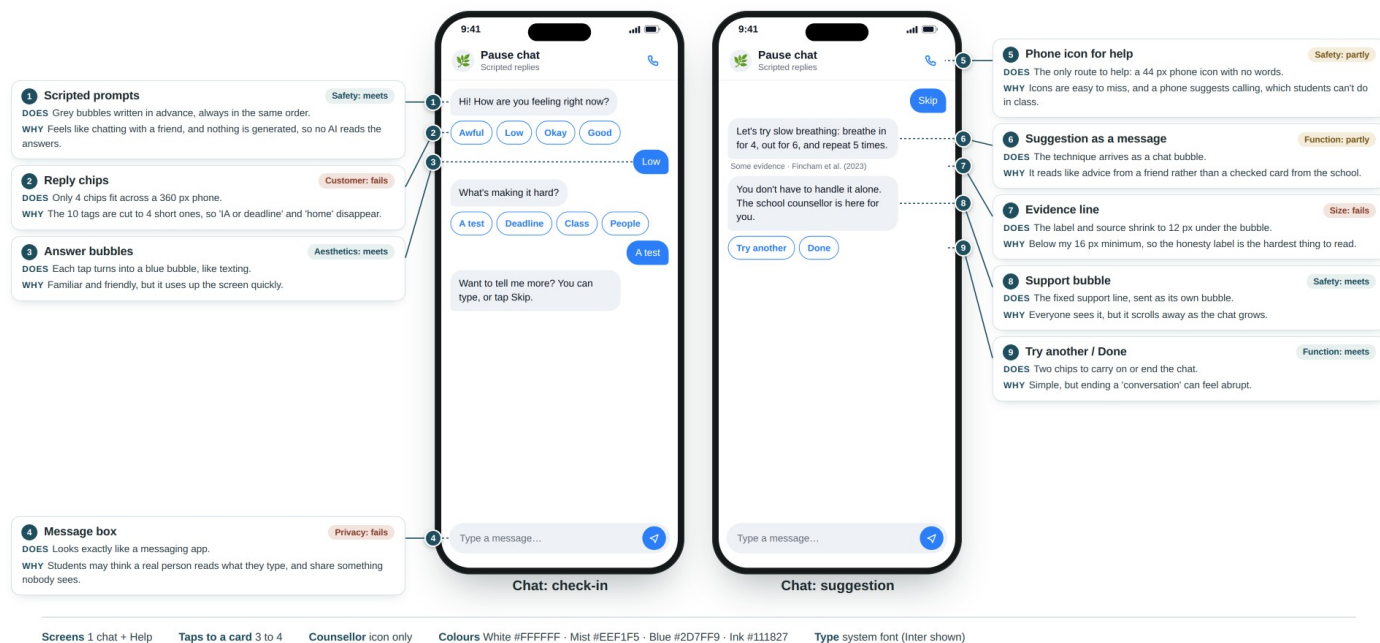


Figure B4: Design 3: Scripted Chat (no AI) (Canva).

ACCESSFMM	How this design meets my B.i specifications	Score
Aesthetics	Clean and consistent, though it looks like any messaging app rather than something designed for calm.	1/1
Cost	Free to build and use.	1/1
Customer / Client	Fails 'fits MYP and DP' because two of the situation tags DP students use most are missing. Chips can be tapped silently, so it is still allowed in class.	1/2
Environment	Text only.	2/2
Safety	The phone icon has no label, so the counsellor specification is only partly met. The message box looks like someone is reading, which only partly meets privacy. The support bubble and the content itself are fine.	2/4 (2 partly)
Size	Fits every screen, but the 12 px evidence line fails 'easy to read'.	1/2
Form and Function	The check-in is spread through a chat, so only partly one screen. 3 to 4 taps is close to the limit but over it, so only partly fast enough. Matching and labels are met.	2/4 (2 partly)
Manufacturing	Buildable in 20 h, but every new card needs its own branch of scripted messages, so only partly easy to extend.	1/2 (1 partly)
Materials	Free and traceable.	1/1
Total	12 of 19 specifications met, 5 partly met, 2 not met.	12/19

Design 4: Check-in Card + Result

Design 4 tries to keep the speed of Design 1 and the calm of Design 2. Everything the student tells the app fits on one check-in card: a mood (the only required part), two short groups of tags and an optional note. One tap on

'Show me something that might help' opens a second screen with a single coping card. The counsellor button sits in the top bar and says 'Talk to the counsellor' in words, in the same place on every screen, like Wysa's always-visible SOS button.

In the same scenario, the student taps 'Bad', reads 'Things feel heavy, but you can still get through the lesson', taps 'anxious' and 'test or exam', then the button. They get 'Worry dump', a writing technique tested with high school students before exams (Ramirez & Beilock, 2011). That is three taps, with the counsellor visible the whole time. The one weakness I found is that on phones shorter than 800 px the submit button drops just below the fold.

Design 4: Check-in Card + Result

Two screens: one check-in card that holds everything, then one result card. Choose a mood, tap a tag, submit. The counsellor button never leaves the top bar.

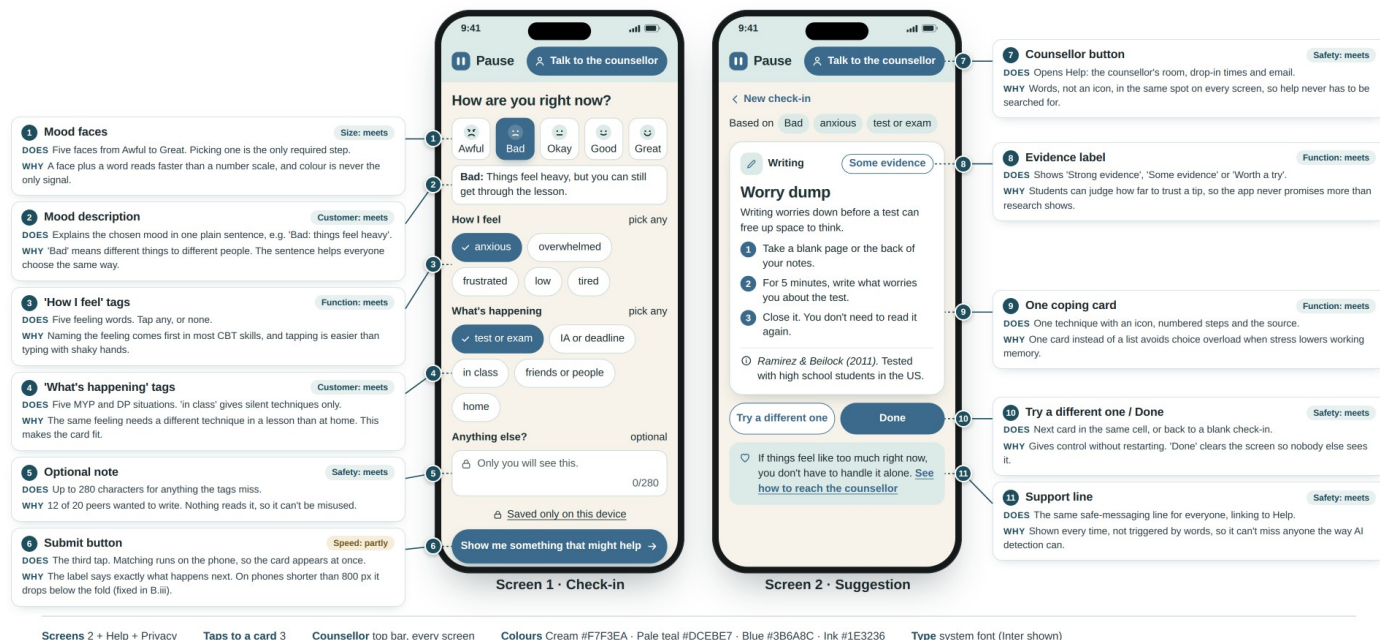


Figure B5: Design 4: Check-in Card + Result (Canva).

ACCESSFMM	How this design meets my B.i specifications	Score
Aesthetics	Cream, pale teal and one blue used the same way on every screen.	1/1
Cost	Free to build and use.	1/1
Customer / Client	All ten tags in two labelled groups, so it fits MYP and DP. 'in class' limits results to silent cards, which the teacher needed.	2/2
Environment	Text only, and matching runs on the phone, so only the first load uses the Wi-Fi.	2/2
Safety	The only design to meet all four. The labelled counsellor button is on every screen, the support line is shown to everyone, history is off by default, and every card is checked by the counsellor.	4/4
Size	Every tap target is at least 44 px and no text is under 16 px.	2/2
Form and Function	Everything is on one check-in screen and the card is matched. Three taps meets the limit, but on short phones the student has to scroll once, so speed is only partly met. Evidence labels on every card.	3/4 (1 partly)
Manufacturing	Two screens that read every word from content.json, so new cards or Arabic need no new screens.	2/2
Materials	Free and traceable.	1/1
Total	18 of 19 specifications met, 1 partly met, 0 not met.	18/19

Can the ideas be understood without me explaining them?

I checked this directly. I gave the four boards to 4 peers who had not seen the project, pointed at a screen, and asked them what each numbered part does. All 4 described every part correctly on Designs 2 and 4, and 3 of 4 did on Design 1. On Design 3, 3 of the 4 thought the message box meant a real person would reply. That misunderstanding is the same safety risk the counsellor later raised (Appendix B3).

iii. Present the chosen design and justify its selection

Out of the four ideas, I chose **Design 4: Check-in Card + Result**. It was not the most popular idea in the peer vote, so I want to be clear about why it won. It meets 18 of my 19 specifications and partly meets the other one, more than any other idea, and it is the only idea that meets all four Safety specifications. Both of my clients, the school counsellor and the teacher, rated it 5 out of 5. In the paper test, students found the counsellor in a median of 2 seconds, against 7 to 14 seconds for the other ideas.

Chosen design: Design 4, Check-in Card + Result

Design 4 meets 18 of the 19 specifications and was the first choice of both clients, the counsellor and the teacher.

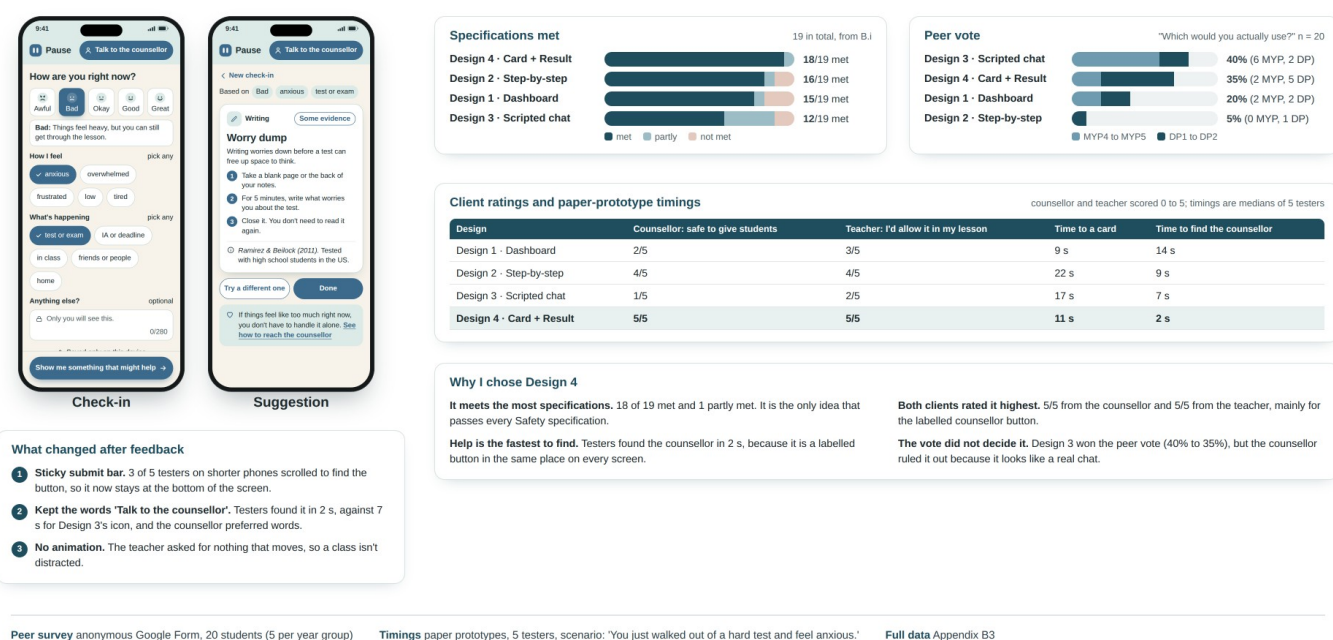


Figure B6: Chosen design: final screens, scores and feedback (Canva).

How I collected feedback

I used four kinds of evidence, so that no single opinion could decide the outcome. First, the specification scores from B.ii. Second, an anonymous Google Form answered by 20 students (5 each from MYP4, MYP5, DP1 and DP2), who saw the four boards in a random order and chose the one they would actually use when stressed at school. Third, a paper-prototype test with 5 testers: I gave them a scenario card ('You just walked out of a hard test and feel anxious') and timed how long it took to reach a card and then to find the counsellor. Finally, my two clients scored the boards from 0 to 5: the counsellor on whether each was safe to give to students, and the teacher on whether they would allow it in a lesson. The raw results are in Appendix B3.

Comparing the four designs

ACCESSFMM	Design 1	Design 2	Design 3	Design 4
Aesthetics	1/1	1/1	1/1	1/1
Cost	1/1	1/1	1/1	1/1
Customer / Client	2/2	2/2	1/2	2/2

ACCESSFMM	Design 1	Design 2	Design 3	Design 4
Environment	2/2	2/2	2/2	2/2
Safety	1/4 (1 partly)	3/4 (1 partly)	2/4 (2 partly)	4/4
Size	1/2	2/2	1/2	2/2
Form and Function	4/4	2/4	2/4 (2 partly)	3/4 (1 partly)
Manufacturing	2/2	2/2	1/2 (1 partly)	2/2
Materials	1/1	1/1	1/1	1/1
Total met	15/19	16/19	12/19	18/19

The totals for Designs 1, 2 and 3 look fairly close, but where they lose marks matters more than how many. Their failures are in Safety and Size, which I treat as non-negotiable for an app used by minors during the school day. A design that hides the counsellor link below the fold is not safe enough, however fast it is. Design 4's only 'partly' is speed on short phones, which can be fixed without changing the idea.

Design	Features and justification	Reason for rejection	Evidence
Design 4 (chosen)	Three taps on one screen; labelled counsellor button on every screen; history off by default; easiest to extend.	N/A	18/19 met. Counsellor 5/5, teacher 5/5. Counsellor found in 2 s.
Design 1	Fastest to a card (2 taps, 9 s).	Counsellor link hidden 1180 px down; tap targets too small; history visible on shared laptops.	15/19 met. Counsellor 2/5. Counsellor found in 14 s, the slowest.
Design 2	Clearest screens and largest targets; my second choice.	5 to 6 taps over three screens; 22 s to reach a card, the slowest.	16/19 met. Counsellor 4/5, teacher 4/5.
Design 3	Friendly and familiar; the peer favourite (40%).	Looks like someone is reading; drops two MYP/DP tags; 12 px evidence label.	12/19 met. Counsellor 1/5.

Why the counsellor's view outweighed the peer vote

Design 3 won the peer vote with 40%, just ahead of Design 4 with 35%. Most of its votes came from MYP students (6 of 8), who said it felt 'like texting a friend'. The counsellor rated it 1 out of 5: 'If it looks like someone is listening, a student might type something serious and wait for a reply that never comes.' I gave the counsellor's view more weight. Criterion A names the counsellor as the person who receives any escalation and decides whether a tool is safe for students, and a design that invites disclosures nobody reads goes against two of my Safety specifications.

The peer vote still mattered. DP students clearly preferred Design 4 (5 of 10 votes), which connects to my related concept of perspective: MYP and DP students want different things from the same app. Design 4 was the one option both age groups and both clients could accept. The teacher put it simply: 'The one with everything on one card is the only one I could see a student finishing before I notice. No moving parts, please.'

Which design is easiest to keep developing

I plan to keep working on Pause after this unit, so I also compared how much work it would take to add more cards or an Arabic version. Design 4 is the easiest: a new card is one entry in content.json and one code in the matching table, and Arabic is a second set of text plus a right-to-left setting, with no new screens. Design 3 would need a new branch of scripted messages for every card, Design 2 gets longer with every new question, and Design 1's page would grow even further below the fold.

What I changed after feedback

- **Sticky submit bar.** 3 of the 5 testers were using shorter phones and had to scroll to find the submit button, so in B.iv the button sits in a bar fixed to the bottom of the check-in screen. This turns Design 4's one 'partly' into 'met'.
- **Kept the words 'Talk to the counsellor'.** Testers found the labelled button in 2 s, against 7 s for Design 3's icon, and the counsellor said: 'Students don't always know what an icon means.'
- **No animation at all.** The teacher said anything that moves would draw attention in a lesson.

Limits of this decision. All of this feedback was on wireframes and paper prototypes, not a working app, and the groups were small (20 students and 5 testers). In Criterion D I will repeat the timing and counsellor-finding tests on the real build.

iv. Develop accurate and detailed planning drawings and outline the requirements for the creation of the chosen solution

This section is my plan for building Pause, written so that someone who has never spoken to me could build the same app. The annotated planning drawings in Canva explain the purpose of each part; the tables here add the behaviour and states that a drawing can't show. All sizes are CSS pixels at 100% on a 360 x 800 px phone. On wider screens (up to 1366 px), the content column stays at most 480 px wide and is centred.

Design tokens (used on every screen)

Cream #F7F3EA	Screen background. Ink text on it: contrast 12.1:1.
Pale teal #DCEBE7	Top bar, tag pills, support box, danger box. Ink text on it: 10.9:1.
Blue #3B6A8C	Buttons, selected chips and moods, links. White text on it: 5.8:1. Blue text on Cream: 5.2:1; on Pale teal: 4.7:1.
Ink #1E3236	All body text and headings.
Border #C9D9D5	1.5 px outlines on chips, mood buttons and boxes.
Type	Device system font stack: -apple-system, 'Segoe UI', Roboto, sans-serif (0 KB download). 24 px bold for titles, 22 px bold for the question, 20 px bold for the wordmark, 16 px for everything else (never smaller).
Spacing and shape	16 px side gutters; 8 px gaps between tap targets; radius 12 to 14 px for boxes and mood buttons, 20 px for the coping card, 22 to 26 px for pill buttons and chips.
Tap targets	Every tappable element is at least 44 x 44 px (W3C, 2024, SC 2.5.5).

Every contrast ratio above passes WCAG AA for normal text (4.5:1 or higher) (W3C, 2024, SC 1.4.3).

Planning drawing 1: Check-in screen

Planning drawing 1: Check-in screen

Final version, with the sticky submit bar added after B.iii feedback. Every size, colour and font needed to rebuild the screen.



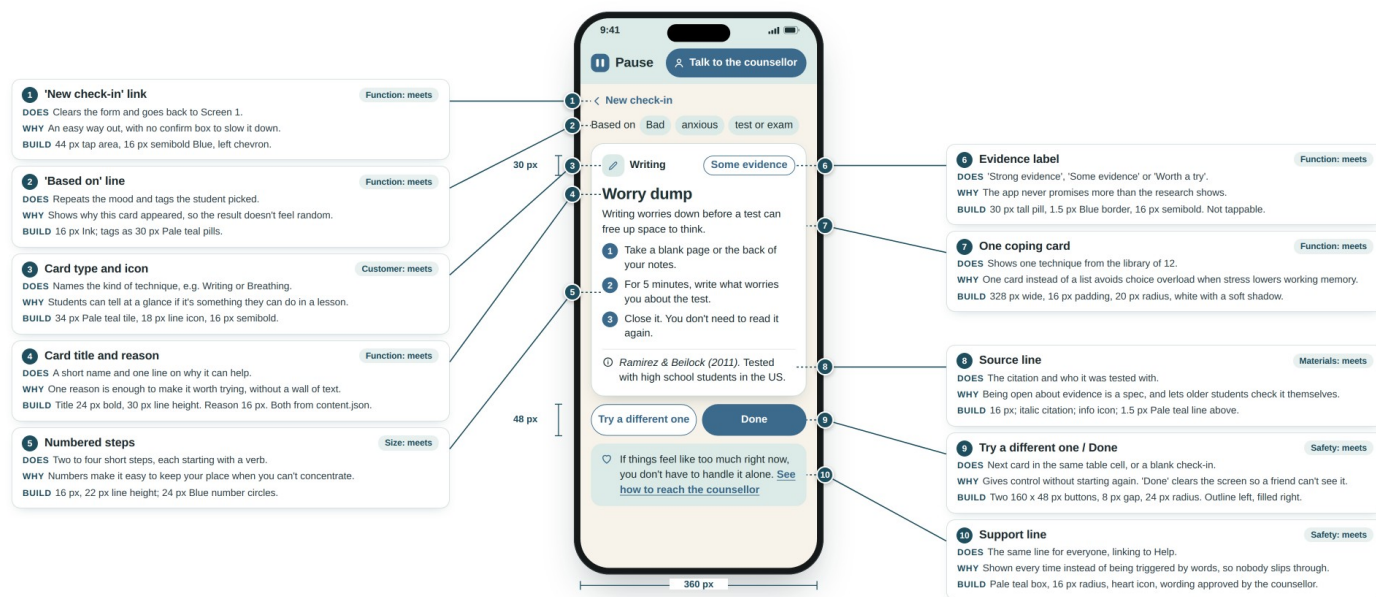
Figure B7: Planning drawing 1, check-in screen (Canva).

Element	Size and position	Colour and type	Behaviour and states
Top bar	360 x 64 px, sticky at the top	Pale teal; wordmark 20 px bold Ink	Shared component on all four screens; never scrolls away.
Counsellor button	About 180 x 44 px, 16 px from the right edge	Blue, white 16 px semibold, 22 px radius	Tap opens Help. On the Help screen it shows a 'you are here' ring.
Question	Full width, 14 px below the bar	22 px bold Ink	Static text from content.json.
Mood buttons	5 x (59 x 64 px), 8 px gaps	White, 1.5 px border, 14 px radius. Selected: Blue, white text	Single choice. Stores 1 to 5. Tapping another mood replaces it. The first choice enables the submit button.
Mood sentence	328 px wide, 8 px below the moods	White box, 12 px radius, 16 px text	Hidden until a mood is picked; shows moods[i].desc from content.json.
Feeling chips	44 px tall, 16 px side padding, wrap	As mood buttons, plus a tick when selected	Multi-select toggle. The first one selected becomes the matching row.
Situation chips	As feeling chips	As above	Multi-select. The first one selected becomes the column; 'in class' limits results to silent cards.
Note box	328 x 70 px	White, 14 px radius, 0/280 counter	Typing stops at 280 characters. Never passed to match(). Saved only if history is on.
Privacy link	Centred, 44 px tap area	16 px underlined Ink, lock icon	Tap opens Privacy.
Sticky submit bar	360 x 80 px; button 328 x 52 px	Cream bar, 1.5 px top line; Blue button	50% opacity and disabled until a mood is picked. Tap runs match() and opens Screen 2.

Planning drawing 2: Suggestion screen

Planning drawing 2: Suggestion screen

One coping card at a time, with its evidence label and source, then the same support line for everyone.



Screen 2 Suggestion, drawn at 90%

Example Bad + anxious + test or exam gives C3 'Worry dump'

Colours Cream #F7F3EA - Pale teal #DCEBE7 - Blue #3B6A8C - Ink #1E3236 - Border #C9D9D5

Type device system font, 16 px minimum (inter shown)

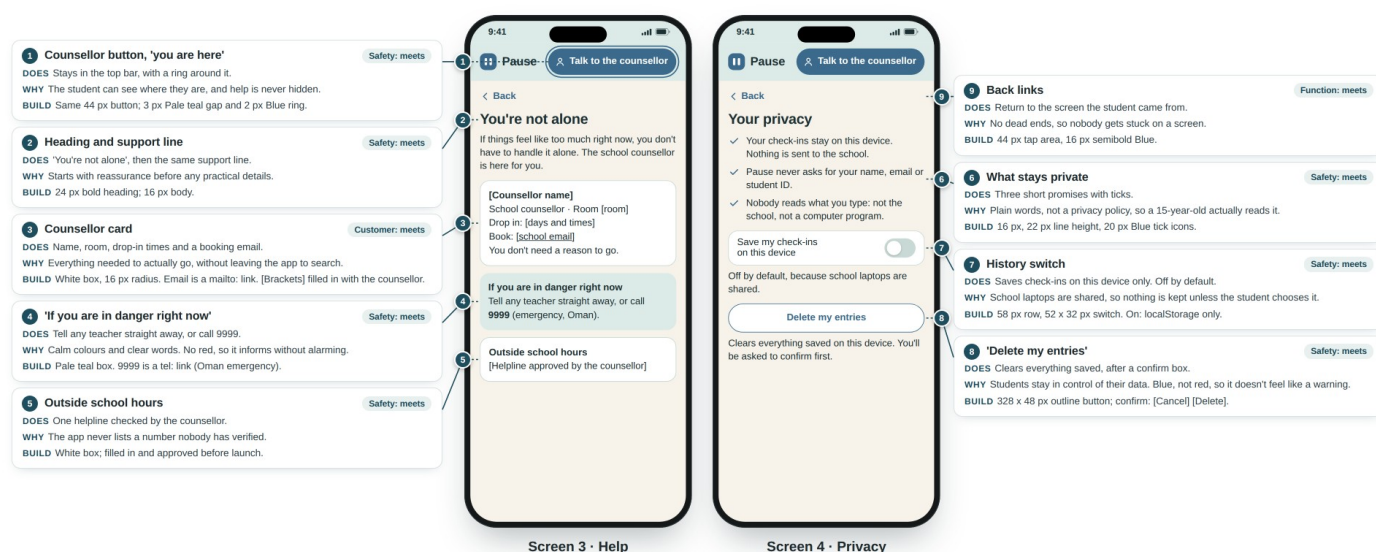
Figure B8: Planning drawing 2, suggestion screen (Canva).

Element	Size and position	Colour and type	Behaviour and states
'New check-in' link	44 px tap area	16 px semibold Blue, chevron	Clears all inputs and returns to Screen 1 with no confirm box.
'Based on' line	Full width	16 px Ink; 30 px Pale teal pills	Shows the mood and the first tag from each group.
Coping card	328 px wide, 16 px padding, 20 px radius	White, 1.5 px border, soft shadow	Filled from content.json using the first card ID that match() returns.
Card type	34 px tile + label	Pale teal tile, 18 px icon, 16 px semibold	Icon and label come from the card's type.
Evidence pill	30 px tall	White, 1.5 px Blue border, 16 px Blue	One of three fixed values; not tappable.
Steps	24 px number circles, 8 px apart	16 px Ink; Blue circles	2 to 4 steps from the card's step list.
Source line	Card width, line above	16 px; italic citation	Citation plus who it was tested with.
Two buttons	2 x (160 x 48 px), 8 px gap	Outline Blue / filled Blue	'Try a different one' moves to the next ID in the list (silent cards only in class). 'Done' clears everything and returns to Screen 1.
Support box	328 px wide, 16 px radius	Pale teal, heart icon, 16 px	Same text for everyone. The link opens Help.

Planning drawing 3: Help and Privacy screens

Planning drawing 3: Help and Privacy screens

Help is one tap away from every screen. Privacy lets the student see and delete everything the app keeps.



Screens 3 and 4 drawn at 82% [Brackets] filled in and checked by the counsellor before launch No red anywhere, even the emergency box Colours Cream #F7F3EA · Pale teal #DCEB7 · Blue #3B6A8C · Ink #1E3236 · Border #C9D9D5

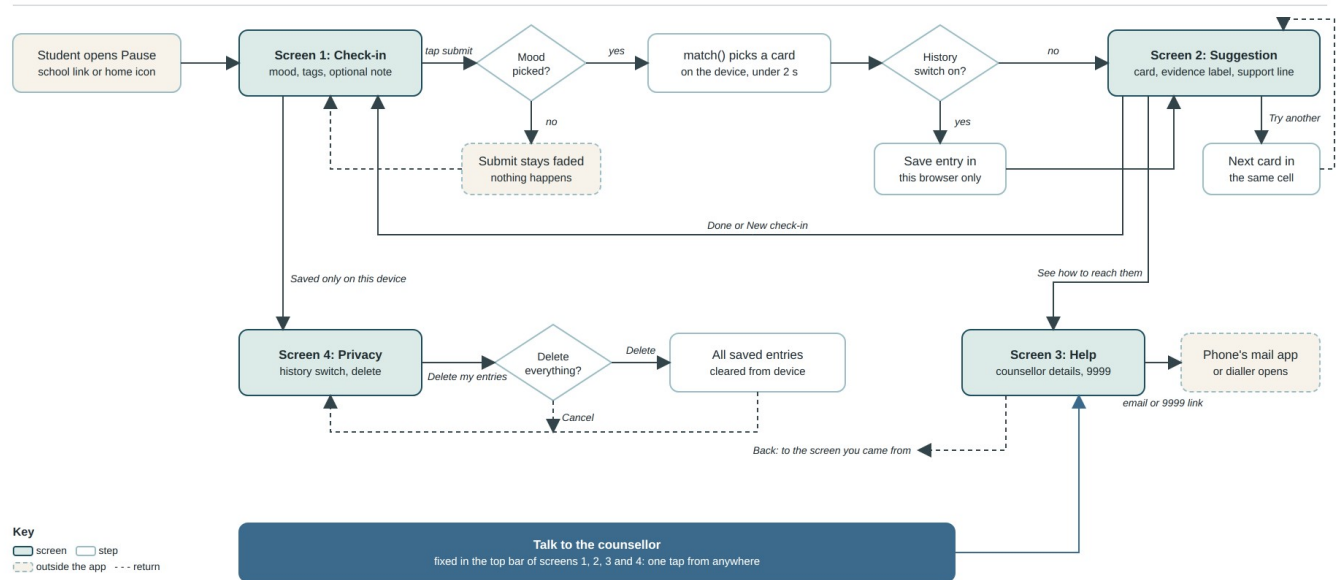
Figure B9: Planning drawing 3, Help and Privacy screens (Canva).

Help: counsellor details	[Counsellor name], room, drop-in times and a booking email as a mailto: link. Filled in and approved by the counsellor before launch.
Help: danger box	Tell any teacher straight away, or call 9999 (emergency, Oman). 9999 is a tel: link. The counsellor confirms the number and wording.
Help: outside school hours	One helpline chosen and checked by the counsellor; nothing unverified is listed.
Support line wording	'If things feel like too much right now, you don't have to handle it alone. The school counsellor is here for you.' Checked against Orygen's #chatsafe guidelines (Orygen, n.d.).
Privacy: switch	Off by default. On: each check-in is saved to pause.entries in this browser only.
Privacy: delete	Opens a confirm box ('Delete all saved check-ins on this device? This can't be undone.') with Cancel and Delete. Delete clears pause.entries.

User flow

User flow

Every screen and every path, including the counsellor button, which reaches Help from all four screens.



Every screen keeps the counsellor button. No path needs a login, a payment or a network request after the first load. Free text is stored only if the history switch is on, and is never read by match().

Figure B10: User flow showing every screen and path (Canva).

Every screen has the counsellor button, so Help is always one tap away. The only required input is the mood. No path asks for a login, a payment or a network request after the first load. Two links leave the app on purpose: the counsellor's email (opens the phone's mail app) and 9999 (opens the dialler).

Matching logic

Matching logic

How the mood and tags become one coping card: a fixed lookup table, checked by hand, with no AI.

- 1 **Read the inputs:** mood (1 to 5, required), the first 'How I feel' tag tapped, and the first 'What's happening' tag tapped. The note is never read.
- 2 **Pick the column:** test or exam / IA or deadline / in class, 'friends or people', 'home' or no tag all use 'Other or none'.
- 3 **Pick the row:** the feeling tag. No feeling tag: mood Awful or Bad uses the 'low' row, anything else uses 'no feeling tag'.
- 4 **Look up the cell:** each cell lists 3 card IDs in order. Show the first one.
- 5 **Safety rule:** if mood = Awful, card R1 'Talk to the counsellor' goes first. The support line shows for everyone anyway.
- 6 **Class rule:** the 'in class' column only holds silent cards (no standing up, talking or phones). This is checked when content.json is saved.
- 7 **Try a different one:** show the next card in the cell, then the rest of the library in order (silent cards only in class), then loop.

Mapping table: row = feeling, column = situation, cell = cards in order

Feeling	Test or exam	IA or deadline	In class (silent only)	Other or none
anxious	C3 Worry dump then C2, B1	C4 One next step then B1, C1	B1 Long out-breath then G2, C2	B2 Double-inhale sigh then G1, C1
overwhelmed	B1 Long out-breath then C3, G1	C4 One next step then M1, B1	G2 Feet on the floor then B1, M2	G1 5-4-3-2-1 then C4, B2
frustrated	B1 Long out-breath then C1, M2	M1 Two-minute reset then C4, B2	B1 Long out-breath then M2, G2	M1 Two-minute reset then B2, C1
low	R1 Talk to the counsellor then G1, C1	M1 Two-minute reset then R2, C4	G2 Feet on the floor then M2, R1	R2 Message someone you trust then M1, R1
tired	M1 Two-minute reset then B2, C2	M1 Two-minute reset then C4, M2	M2 Look away then B1, G2	M1 Two-minute reset then M2, B2
no feeling tag	B1 Long out-breath then C2, G1	C4 One next step then M1, B1	B1 Long out-breath then G2, M2	M1 Two-minute reset then G1, B1

The 5 test inputs from B.i (Form and Function: a matched coping suggestion)

Test	Mood	Feeling	Situation	First card shown	Why
Test 1	Bad	anxious	test or exam	C3 Worry dump	anxious row, Test column
Test 2	Okay	overwhelmed	IA or deadline	C4 One next step	overwhelmed row, IA column
Test 3	Bad	anxious	in class	B1 Long out-breath	In class column, so silent cards only
Test 4	Awful	low	home	R1 Talk to the counsellor	mood Awful puts R1 first (safety rule)
Test 5	Good	(none)	(none)	M1 Two-minute reset	no tags, so 'no feeling tag' row and 'Other' column

Card library (12 cards): B = breathing, G = grounding, C = CBT reframing, M = movement or break, R = reaching out

B1 Long out-breath (Some evidence) silent

B2 Double-inhale sigh (Some evidence) silent

G1 5-4-3-2-1 (Worth a try) silent

G2 Feet on the floor (Worth a try) silent

C1 Check the thought (Some evidence) silent

C2 Nerves can help (Some evidence) silent

C3 Worry dump (Some evidence) silent

C4 One next step (Worth a try) silent

M1 Two-minute reset (Some evidence)

M2 Look away (Worth a try) silent

R1 Talk to the counsellor (Worth a try) silent

R2 Message someone you trust (Worth a try)

Rule-based lookup only: no AI and no free-text reading. Every possible input has a fixed, checked output. Full card text and sources are in the B.iv card library table.

Figure B11: Matching logic, mapping table and the five test inputs (Canva).

The rules in Figure B11, written so they can be turned straight into code:

```
function match(mood, feeling, situation):
    col = {'test or exam': 'Test', 'IA or deadline': 'IA', 'in class': 'Class'}[situation] or 'Other'
    row = feeling if feeling else ('low' if mood <= 2 else 'no feeling tag')
    cards = TABLE[row][col] # 3 card IDs in order
    if mood == 1 and cards[0] != 'R1': cards = ['R1'] + cards # Awful: counsellor first
    rest = [c for c in LIBRARY if c not in cards and (col != 'Class' or c.silent)]
    return cards + rest # 'Try a different one' walks this list
```

Full card library. Every possible input leads to one of these 12 cards. The label follows three rules: Strong evidence = backed by systematic reviews or clinical guidelines for this kind of use; Some evidence = backed by trials, but small, mixed, or not tested with teenagers; Worth a try = low risk and widely recommended, with little direct research. No card earned 'Strong evidence'. The strong reviews tested full courses or regular habits, not one technique used for a minute in a school day, so I did not give that label to any card.

Card	Type	Steps shown	Reason shown	Label, source and who it was tested with
B1: Long out-breath	Breathing Silent: yes	1. Breathe in through your nose for a count of 4. 2. Breathe out slowly through your mouth for a count of 6. 3. Repeat 5 times (about 1 minute).	A longer out-breath can help your body slow down.	Some evidence Fincham et al. (2023) Meta-analysis of trials with adults; small to medium effects, and the authors ask for better-quality trials.

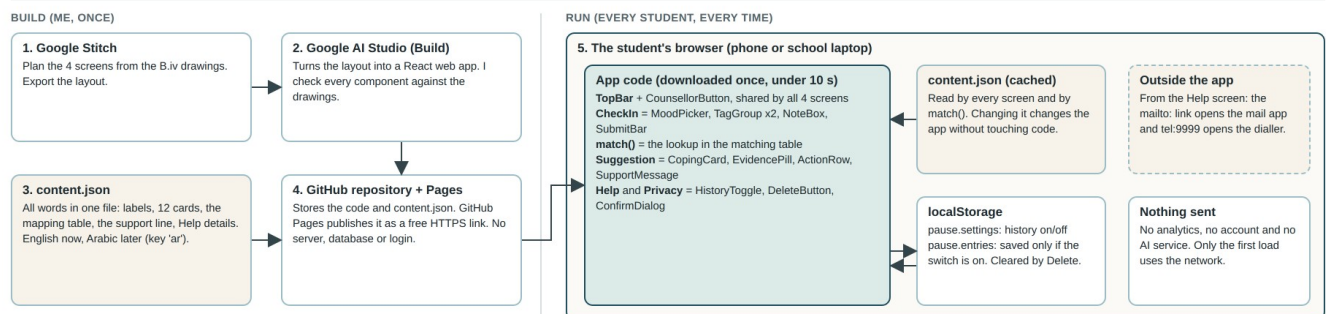
Card	Type	Steps shown	Reason shown	Label, source and who it was tested with
B2: Double-inhale sigh	Breathing Silent: no	1. Breathe in through your nose. 2. At the top, take one more short sip of air. 3. Breathe out slowly and fully through your mouth. 4. Repeat for 1 to 5 minutes.	In one trial, breathing with a long out-breath lifted mood more than mindfulness did.	Some evidence Balban et al. (2023) One month-long online trial with adults.
G1: 5-4-3-2-1	Grounding Silent: yes	1. Name 5 things you can see. 2. Name 4 things you can feel and 3 you can hear. 3. Name 2 things you can smell and 1 you can taste.	Grounding is one of the WHO's self-help skills. This version uses your senses.	Worth a try World Health Organization (2020) Recommended in a WHO self-help guide; this exact version has not been tested on its own.
G2: Feet on the floor	Grounding Silent: yes	1. Slowly push your feet into the floor. 2. Press your palms together or onto the desk. 3. Notice the chair holding you up. 4. Look around and name where you are.	This is the grounding exercise from the WHO's stress guide.	Worth a try World Health Organization (2020) Recommended in a WHO self-help guide; not tested on its own with teenagers.
C1: Check the thought	CBT reframing Silent: yes	1. Write the worrying thought in one line. 2. Ask: what is the evidence for it, and against it? 3. Ask: what would I tell a friend who thought this? 4. Write a fairer version of the thought.	Questioning a thought is a core skill in CBT.	Some evidence Hofmann et al. (2012) CBT was tested as full courses, usually with a therapist, not as a single card.
C2: Nerves can help	CBT reframing Silent: yes	1. Notice the racing heart or butterflies. 2. Tell yourself: 'This is my body getting ready.' 3. Take one slow breath and start.	Students told that nerves can help them did better on a practice test.	Some evidence Jamieson et al. (2010) One study with US students preparing for the GRE.
C3: Worry dump	CBT reframing (writing) Silent: yes	1. Take a blank page or the back of your notes. 2. For 5 minutes, write what worries you about the test. 3. Close it. You don't need to read it again.	Writing worries down before a test can free up space to think.	Some evidence Ramirez & Beilock (2011) Tested with high school and university students in the US.
C4: One next step	Problem solving Silent: yes	1. Write the whole task in one line. 2. Split it into 3 small pieces. 3. Do only the first piece for 10 minutes.	Problem-solving skills help protect teenagers' mental health.	Worth a try World Health Organization (2024) The WHO lists problem-solving skills as protective; this exact method was not tested.
M1: Two-minute reset	Movement or break Silent: no	1. Stand up and stretch your arms and shoulders. 2. Walk to refill your water bottle. 3. Come back and start with the easiest part.	Short breaks raised energy and lowered tiredness across 22 studies.	Some evidence Albulescu et al. (2022) Mostly adults at work or in lab tasks; small effects.

Card	Type	Steps shown	Reason shown	Label, source and who it was tested with
M2: Look away	Movement or break Silent: yes	1. Look at something far away for 20 seconds. 2. Roll your shoulders back twice. 3. Unclench your jaw and hands.	Even a very short pause can lower tiredness.	Worth a try Albulescu et al. (2022) Based on the micro-break review; this exact version was not tested.
R1: Talk to the counsellor	Reaching out Silent: yes	1. Tap 'Talk to the counsellor' at the top. 2. See when you can drop in, or send a booking email. 3. You don't need a reason to go.	Support at school is one of the things that protects teenagers' mental health.	Worth a try World Health Organization (2024) General WHO guidance, not a trial of this app.
R2: Message someone you trust	Reaching out Silent: no	1. Pick one person: a friend, a parent or another adult you trust. 2. Send one line, like 'Rough day, can we talk later?' 3. You don't have to explain everything.	Good relationships with family and friends help protect teenagers' mental health.	Worth a try World Health Organization (2024) General WHO guidance, not a trial of this app.

How the parts connect

How the parts connect

What I build once, what runs on each student's device, and where the data stays.



How it works, step by step

- Student opens the GitHub Pages link (or a home-screen icon added from it). The browser downloads the app and content.json once.
- Screen 1 shows the counsellor button, 5 mood buttons, 10 tags and the note box. Nothing is asked before this screen.
- The student picks a mood (required) and any tags, then taps the submit button: 3 taps at the least.
- match() reads the mood and the first tag in each group, finds the cell in the mapping table, and returns 3 card IDs.
- Screen 2 draws the first card from content.json: label, title, reason, steps and source, plus the support line.
- If the history switch is on, the entry is saved in localStorage. If not, nothing is kept after 'Done'.
- 'Try a different one' shows the next card. 'Done' goes back to a blank Screen 1.
- From any screen, the counsellor button opens Help, which links to the counsellor's email and to 9999.

Software: Google Stitch, Google AI Studio, GitHub, GitHub Pages (all free). Storage: on the device only (localStorage). Network: only the first load needs school Wi-Fi.

Figure B12: System diagram and step-by-step operation (Canva).

Component	Contains and does
TopBar	Wordmark + CounsellorButton. Used on all 4 screens.
CheckIn	MoodPicker (5 buttons), TagGroup x 2 (feelings, situations), NoteBox, PrivacyLink, SubmitBar.
match()	Pure function: (mood, firstFeeling, firstSituation) to a list of card IDs. No network, no AI.
Suggestion	BasedOnline, CopingCard (EvidencePill, title, reason, steps, source), ActionRow (Try a different one, Done), SupportMessage.

Component	Contains and does
Help	BackLink, support text, CounsellorCard, DangerBox, OutsideHoursBox.
Privacy	BackLink, PrivacyList, HistoryToggle, DeleteButton, ConfirmDialog.
storage.js	load(), save(entry), clear(). Uses localStorage keys pause.settings and pause.entries only.

Data stored on the device. Nothing is sent anywhere. With history off (the default), nothing is kept after 'Done'.

```
pause.settings = { "historyOn": false, "lang": "en" }
pause.entries = [ { "time": "2026-11-02T10:41", "mood": 2, "feelings": ["anxious"],
                    "situations": ["test or exam"], "note": "", "cardShown": "C3" } ] // only if
historyOn
```

content.json structure. All words live here, so new cards or an Arabic version need no new screens (Manufacturing: easy to extend later).

```
{ "en": { "moods": ["Awful", "Bad", "Okay", "Good", "Great"],
          "feelings": ["anxious", "overwhelmed", "frustrated", "low", "tired"],
          "situations": ["test or exam", "IA or deadline", "in class", "friends or people", "home"],
          "support": "If things feel like too much right now, ...",
          "help": { "name": "[Counsellor name]", "room": "[room]", "email": "[school email]" },
          "cards": { "C3": { "name": "Worry dump", "label": "Some evidence", "steps": [...],
                           "why": "...", "source": "Ramirez & Beilock (2011)", "silent": true } } },
  "ar": { ... same keys, Arabic text ... },
  "table": { "anxious": { "Test": ["C3", "C2", "B1"], "IA": [...], "Class": [...], "Other": [...] } } }
```

How I will build it, step by step

1. Plan screens (2 h)	In Google Stitch, recreate the 4 screens from Planning drawings 1 to 3 using the design tokens above. Export to AI Studio.
2. Generate the app (4 h)	In Google AI Studio (Build), create a React app with the 4 screens and the TopBar shared between them. Check every size and colour against the drawings with the browser's inspect tool.
3. Add content (3 h)	Write content.json from the card library table and the mapping table. Make every screen read its words from it.
4. Add match() (2 h)	Write match() from the pseudocode. Test it with the 5 test inputs in Figure B11; each must return the listed first card.
5. Add storage and Privacy (2 h)	Write storage.js, the history switch (off by default) and Delete with its confirm box.
6. Counsellor sign-off (1 h)	Show the counsellor the Help screen, support line and all 12 cards. Make their changes in content.json.
7. Publish (1 h)	Push the code to a GitHub repository and turn on GitHub Pages to get a free HTTPS link.
8. Test against B.i (5 h)	Run every B.i test: 3 school devices, 320/375/768/1366 px widths, contrast checks, font size check (0 KB), timing with 5 peers, privacy test with the word TESTPRIVACY.

Total: 20 hours, which matches Manufacturing: built in a browser in about 20 hours.

Requirements for creating the solution

Software	Google Stitch and Google AI Studio (free, in the browser); a code editor if needed (VS Code, free); Chrome, Safari and Edge for testing.
Hardware	A laptop for building; 3 school devices (a Windows laptop, a Chromebook or Mac, and one phone) for testing.

Internet and connectivity	Internet to build and publish. Students only need school Wi-Fi for the first load; after that, the check-in and matching run on the device.
Accounts and platforms	A Google account (Stitch and AI Studio) and a GitHub account (repository and Pages). Students need no account at all.
Tools	Browser inspect tool and Network tab; the WebAIM Contrast Checker; a stopwatch; the B.i test sheets.
Equipment	Printed paper prototypes and scenario cards for testing; nothing physical is part of the product.
Materials and resources	content.json written from my sources; free-to-use face icons drawn as simple SVG; an asset register recording the source and licence of everything.
Skills and knowledge	Basic React and JSON editing; reading and checking AI-generated code; using inspect tools; writing safe-messaging wording; APA referencing for the card sources.
Technical requirements	Runs in any modern browser on iOS, Android, Windows, macOS and ChromeOS; 320 to 1366 px wide; loads in 10 s or less on school Wi-Fi; no sounds, notifications or animation; WCAG AA contrast and 44 px targets.
Research and resources	My B.i specifications; the card sources in the reference list; Orygen's #chatsafe guidelines for the support wording; WCAG 2.2.
Special conditions	The counsellor must approve the Help details, the support line and all 12 cards before launch. The school must allow the link on school Wi-Fi and agree that students can open it in lessons. Pause is a signpost to people, not a replacement for the counsellor, and it never contains supplement or medication content.

Final check: could someone else build Pause from these plans?

Every screen drawn with sizes	Yes: Planning drawings 1 to 3 and their element tables.
Every colour, font and size	Yes: Design tokens table.
Every path between screens	Yes: User flow (Figure B10).
What every mood and tag means	Yes: Mood and tag guide (Figure B1) and content.json.
How inputs become a suggestion	Yes: Matching logic, pseudocode and the full card library.
Where data is stored	Yes: storage keys and the content.json structure.
Tools, accounts and build order	Yes: build steps and requirements tables.
Who must approve what	Yes: Special conditions (counsellor sign-off, school permission).

Appendix B3: Feedback on the design ideas

Peer vote (n = 20)	Design 3: 8 (6 MYP, 2 DP). Design 4: 7 (2 MYP, 5 DP). Design 1: 4 (2 MYP, 2 DP). Design 2: 1 (0 MYP, 1 DP). Most common reasons: Design 3 'feels like texting a friend' (6); Design 4 'quick and nothing extra' (5); Design 1 'everything on one page' (3).
Paper-prototype test (n = 5)	Median time to reach a card: D1 9 s, D2 22 s, D3 17 s, D4 11 s. Median time to find how to contact the counsellor: D1 14 s, D2 9 s, D3 7 s, D4 2 s. 3 of 5 testers on shorter phones had to scroll to find Design 4's submit button.
Counsellor	Ratings for 'Is this safe to give to students?': D1 2, D2 4, D3 1, D4 5. 'The chat one worries me. If it looks like someone is listening, a student might type something serious and wait for a reply that never comes.' 'I like that the button says counsellor. Students don't always know what an icon means.'
Clarity check (n = 4)	Peers who had not seen the project explained each numbered part of each board. Correct for every part: Design 1, 3 of 4; Design 2, 4 of 4; Design 3, 1 of 4 (3 thought the message box meant a real person would reply); Design 4, 4 of 4.
Teacher	Ratings for 'Would you allow this in your lesson?': D1 3, D2 4, D3 2, D4 5. 'The one with everything on one card is the only one I could see a student finishing before I notice. No moving parts, please.'

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